Exploring DPO's Implicit Reward

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Abstract

Direct Preference Optimization (DPO) claims to skip the reward-model stage of RLHF entirely: a DPO-trained policy is itself, implicitly, a reward model. We test this claim as a preference-ranking task. Under a matched training budget we train a Qwen2.5-0.5B policy with LoRA from one shared SFT checkpoint into three scorers (a DPO policy's implicit reward, an explicitly trained Bradley–Terry reward model, and an SFT log-probability baseline with no reward-specific training), and score all three as pairwise classifiers in distribution, on four RewardBench shift subsets, and on a cross-dataset shift set, over five seeds. In distribution the explicit reward model wins (0.663 vs. 0.600 length-controlled accuracy, all five seed-level McNemar tests on the full test set significant): **H1 is supported**. The predicted widening under shift does not occur; where a difference is resolvable (two of five shifted sets) it favors the implicit reward: **H2 is not supported**. The SFT baseline beats both trained scorers on RewardBench-Reasoning, and the implicit reward on two further shifted sets. A four-epoch run confirms the in-distribution ranking is not an artifact of stopping early, and a surface-feature analysis implicates response length in part of that gap, though the explicit model's length bias fades, not grows, as it trains.

1 Introduction

Reinforcement Learning from Human Feedback (RLHF) is the standard recipe for aligning a pre-trained language model with human preferences: collect pairwise comparisons between candidate responses, train an explicit reward model to predict which one people preferred, then optimize the policy against that learned reward with reinforcement learning (typically PPO) under a KL penalty holding the policy near a reference model [1, 2]. That pipeline needs a separately trained reward model and an unstable, hyperparameter-sensitive RL stage. Direct Preference Optimization (DPO) [3] skips both, training a policy directly on preference pairs with a simple supervised-style loss while implicitly optimizing the same KL-regularized objective. Its title makes the strong claim: a DPO-trained policy *is*, secretly, also a reward model. That reward is recoverable as $\hat{r}(x, y) = \beta \log[\pi_\theta(y \mid x)/\pi_{\text{ref}}(y \mid x)]$, but the equivalence is exact only at the optimum, which a trained policy need not reach. What we test is $\hat{r}$ as a practical preference scorer, in and out of distribution, not the theorem itself.

Christiano et al. [1] introduced the core idea: learn a reward function from pairwise human preferences and optimize a policy against it with RL. Its strength is generality, since only comparative feedback is needed. But routing through PPO leaves the pipeline sensitive to reward-model quality and prone to reward hacking under prolonged optimization. Ouyang et al. [2] scaled this to language models with InstructGPT, training a Bradley–Terry-style reward model [4] and fine-tuning the policy against it. The alignment gains were large, but the paper documents the same PPO instability and adds the cost of serving a separate reward model. Gao et al. [5] quantify how far a learned reward can be optimized before proxy and true reward diverge, the failure our held-out comparison probes, although their scaling laws are fitted against synthetic gold reward models, not human labels. DPO [3] removes that stage with a simpler, more stable loss on the preference pairs themselves. Its limitation is that it discards the reward model as an explicit, separately-evaluable object, so reuse of $\hat{r}$ as a scorer has never been validated the way an explicit reward model would be, and we found no prior work comparing the two directly under matched training conditions. SimPO [6] is one of several post-DPO variants that drop the reference policy for a length-normalized reward, saving memory and curbing length bias, but in doing so it removes the very $\pi_\theta/\pi_{\text{ref}}$ ratio whose reuse as a reward model we test here. Closest to a stress test is RewardBench [7], built around adversarial, distribution-shifted comparisons, but it evaluates only explicitly-trained reward models, never a policy read out as an implicit one. Three further ingredients make the question tractable at small scale. LoRA [8] lets several full training conditions run on one consumer GPU, at the cost of not testing full-parameter fine-tuning. UltraFeedback [9] supplies the preference pool our budget-matched training draws from, with the caveat that its labels come from an AI judge. Anthropic's HH-RLHF [10], our cross-dataset shift set, was human-annotated instead, so it tests generalization across the labelling process as well as the data.

This gap matters practically. If the implicit reward generalizes as well as an explicit one, DPO checkpoints could be reused as reward models, removing a stage from the RLHF pipeline; if not, the "secretly a reward model" framing needs qualifying as holding only in a narrow, training-distribution sense. We test two hypotheses stated in advance in our project proposal: **H1**, the explicit reward model beats the implicit reward on held-out in-distribution pairs, and **H2**, that gap widens under distribution shift. We committed beforehand to reporting either outcome, since a null result would itself substantiate DPO's claim at small scale.

1.1 Our Contribution

- **Budget-matched pipeline:** the DPO policy and the explicit reward model differ only in their loss (Figure 1).
- **Correctness pass:** four silent bugs found and fixed, and the implicit-reward extraction verified against the DPO trainer's own computation (§3.2).
- **Evaluation:** both scorers, plus an SFT log-probability baseline, scored as pairwise classifiers across six test sets over five seeds (§2.4).
- **Ablations:** a four-epoch extended-training run and a per-checkpoint length-bias-drift analysis (§3.7).

2 Material and Methods

2.1 Overview

Figure 1 shows the pipeline. One supervised fine-tuning (SFT) pass produces the shared reference policy π_{ref}. Two conditions branch off that identical starting point: DPO produces a policy π_θ whose implicit reward is read off after training, with no reward-specific training of its own, and a Bradley–Terry reward model is trained explicitly on the same pairs. Both are then evaluated as pairwise preference classifiers on held-out data.

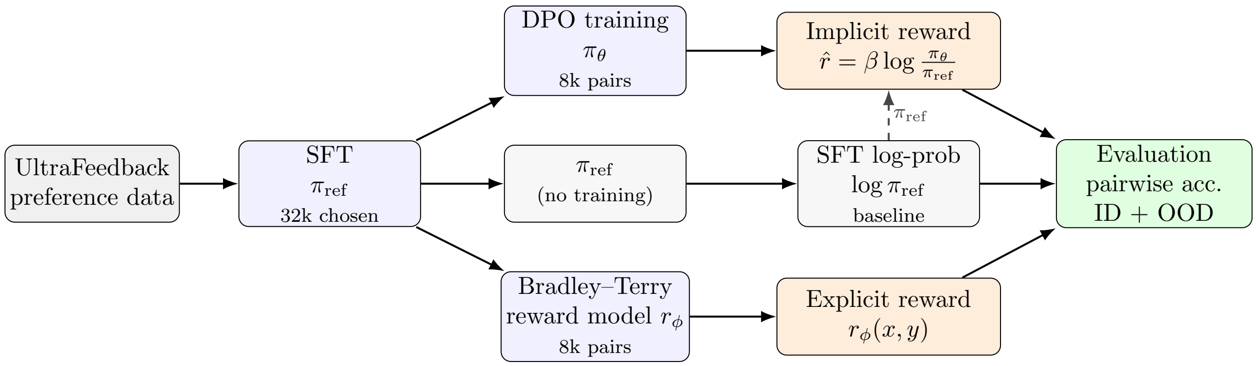

Figure 1: Pipeline overview (left to right). **(1)** One SFT pass on 32k *chosen* UltraFeedback responses yields the shared reference policy π_{ref}. **(2)** From that identical start, two conditions train on the *same* 8k preference pairs, differing only in loss: DPO ($\to \pi_\theta$) and a Bradley–Terry reward model ($\to r_\phi$); π_{ref} is also kept untrained. **(3)** Three scorers are read off: the implicit reward $\hat{r} = \beta \log(\pi_\theta/\pi_{\text{ref}})$, which reuses π_{ref} (dashed); the explicit reward r_ϕ; and the SFT log-probability baseline $\log \pi_{\text{ref}}$. **(4)** All three are evaluated as pairwise preference classifiers, in distribution and under shift.

2.2 Dataset Description

All data are established public benchmarks (Table 1). UltraFeedback [9] supplies the training pairs and the in-distribution held-out test set. RewardBench [7] supplies four shift subsets (Chat, Chat-Hard, Safety, Reasoning) built to stress-test reward models on harder and adversarially-constructed comparisons. Anthropic's HH-RLHF [10] (the test split of `harmless-base`) supplies a cross-dataset shift test: a different collection process entirely, with human rather than AI-judge labels. Every example is normalized to the same (*prompt*, *chosen*, *rejected*) schema. HH-RLHF needs one extra step, since its transcripts are multi-turn and the branches differ only in the final assistant turn: everything up to the last `Human:` turn becomes the prompt, and that final turn the response. Training pairs are drawn from a common length-eligible pool so both conditions see the identical first 8,000 examples (Table 4 says why that pool is necessary).

Table 1: Evaluation sets. "Length-matched" counts the pairs surviving the 1.2 length-ratio filter (§2.4); the three marked † fall below 100 (§3.8).

Set	Role	Distribution	Pairs	Length-matched
UltraFeedback	training + ID test	in-distribution	1,997	442
RewardBench: Chat	OOD test	distribution shift	358	32†
RewardBench: Chat-Hard	OOD test	distribution shift	456	62†
RewardBench: Safety	OOD test	distribution shift	740	83†
RewardBench: Reasoning	OOD test	distribution shift	1,431	908
HH-RLHF (harmless)	OOD test	cross-dataset shift	2,312	276

2.3 Tools, Models, and Algorithms Used

The base model for every condition is Qwen2.5-0.5B [11], a small, publicly available pretrained language model. It is the base checkpoint, not the instruction-tuned one; supplying that stage is what the shared SFT pass is for. It is small enough that all three conditions plus the extended-training ablation fit on one 24 GB GPU (NVIDIA A10G). All fine-tuning uses Low-Rank Adaptation (LoRA) [8] with rank $r = 16$, $\alpha = 32$, dropout 0.05, applied to every attention and MLP projection matrix, via HuggingFace's TRL library. The shared reference policy π_{ref} is one epoch of SFT on the *chosen* responses from 32,000 length-eligible UltraFeedback pairs (1,998 optimizer steps); as a shared input to the matched comparison rather than one of the compared conditions, it is not budget-limited. Figure 2 summarizes the three training objectives.

(a) SFT
Objective: next-token cross-entropy on the *chosen* response only.
Produces: π_{ref}, merged into standalone weights (*not* left as an adapter; see §3.2).

(b) DPO
Loss: $-\log \sigma\big(\beta \log \frac{\pi_\theta(y_c|x)}{\pi_{\text{ref}}(y_c|x)} - \beta \log \frac{\pi_\theta(y_r|x)}{\pi_{\text{ref}}(y_r|x)}\big)$
Produces: π_θ, a fresh zero-init LoRA on top of π_{ref}; the reward is read off *after* training, not trained on directly.

(c) Reward model
Loss: $-\log \sigma\big(r_\phi(x, y_c) - r_\phi(x, y_r)\big)$ (Bradley–Terry loss [4])
Produces: r_ϕ, a scalar head on the same π_{ref} backbone, trained on purpose to predict the winner.

Figure 2: The three training objectives, all starting from the identical π_{ref} checkpoint produced by (a).

Table 2 lists the hyperparameters shared across SFT, DPO, and the reward model; the only quantities that differ between the two conditions are the loss function itself and, for the extended-training ablation, the number of epochs.

Table 2: Shared hyperparameters; $\beta = 0.1$ is DPO's implicit-reward temperature.

Base model	Qwen2.5-0.5B
LoRA rank / α / dropout	16 / 32 / 0.05
LoRA target modules	q,k,v,o,gate,up,down projections
Effective batch size	16 (batch 4 $\times$ grad. accum. 4)
Learning rate (SFT / DPO & RM)	2×10^{-4} / 5×10^{-5}, cosine schedule
Precision	bf16 (single NVIDIA A10G)
Training pairs (headline run)	8,000, budget-matched, 1 epoch
DPO β (headline run)	0.1
Seeds (headline run)	5
Extended-training ablation	4 epochs, 2 seeds, held-out eval every 100 steps

2.4 Performance Evaluation

Every trained scorer produces a scalar $s(x, y)$: for the implicit reward, $s = \hat{r} = \beta \log[\pi_\theta(y \mid x)/\pi_{\text{ref}}(y \mid x)]$; for the reward model, $s = r_\phi(x, y)$; for the SFT baseline, $s = \log \pi_{\text{ref}}(y \mid x)$. Each s is used only for ranking (the scorer's decision is the sign of $s(x, y_c) - s(x, y_r)$), and *pairwise accuracy*, the fraction of pairs ranked correctly, is the primary outcome.

Length-controlled accuracy restricts this to pairs with $\max(L_c, L_r)/\max(1, \min(L_c, L_r)) \leq 1.2$, where L_c, L_r are character lengths of the response text (prompt excluded, un-tokenized); the filter is aggressive on the smaller RewardBench subsets. We also report non-neural floors (random, pick-longer, TF-IDF+LR) alongside the SFT baseline.

ECE is computed over 10 equal-width confidence bins, per seed on the full test set (not length-controlled) and averaged. *McNemar's exact test* (a binomial test on the discordant pairs) is run per seed between the implicit and explicit scorers, also on the full test set, so it tests *accuracy* and not length-controlled accuracy; the two can disagree, and on one set below they do. We report the range of the five p-values, not their mean, since independent p-values should not be averaged, and read each at $\alpha = 0.05$ with no correction for the number of seeds or sets. The reported 95% CI is $\bar{x} \pm t_{0.975,4}\, s/\sqrt{5}$ over the five seed-level values of a given (test set, scorer, metric) triple, where s is their sample standard deviation (§3.8 discusses what this interval does and does not capture). We write *CI excludes/includes 0* for whether a paired difference is *resolvable* at this sample size: explicit minus implicit in Table 5, or either scorer minus the SFT baseline in §3.5.

Data splits. All training draws on UltraFeedback's `train_prefs`: the 8,000 budget-matched pairs for DPO and the reward model, and the 32,000 behind the shared SFT reference. All evaluation draws on data no condition trained on: `test_prefs` (1,997 pairs, after dropping empty responses) for the ID test, plus RewardBench and HH-RLHF, untouched by training. There is no separate validation split. The 400-pair slice tracking held-out accuracy during the extended-training run (§3.7) and the 499-pair slice behind the bias-drift analysis both come from `test_prefs`, so both are subsets of the ID test set. Nothing was tuned on them, but the four-epoch curves are consequently measured on the same distribution, and partly the same examples, as the ID column of Table 5.

3 Experimental Analysis

Code, configuration files, and the full pipeline are available at `https://github.com/silvererudite/dpo-implicit-reward`, and the trained adapters and per-seed checkpoints at `https://huggingface.co/Shamima/dpo-implicit-reward`. Every per-pair score is persisted to disk, so every accuracy, calibration, and significance number below is recomputable without re-running any model. Optimizer-step counts and wall-clock timings are the exception: those come from training logs we did not commit.

3.1 Non-neural reference floors

Table 3 shows the zero-training reference floors. Pick-longer wins RewardBench-Chat (0.805) but collapses on Chat-Hard (0.294), which inverts that cue by construction, so raw accuracy here is substantially a length signal. This motivates length-controlled accuracy as the primary metric below. TF-IDF+LR lands below chance on Reasoning (0.282), suggesting the lexical cues it relies on are actively misleading there.

Table 3: Non-neural reference floors (pairwise accuracy, no language-model training; TF-IDF+LR does fit a logistic regression). Pick-shorter is omitted as it is exactly 1−pick-longer.

Test set	Random	Pick-longer	TF-IDF + LR
UltraFeedback (ID)	0.500	0.568	0.618
RewardBench: Chat	0.500	0.805	0.754
RewardBench: Chat-Hard	0.500	0.294	0.382
RewardBench: Safety	0.500	0.420	0.450
RewardBench: Reasoning	0.500	0.397	0.282
HH-RLHF (harmless)	0.500	0.445	0.453

3.2 Pipeline correctness pass

We cross-checked our implicit-reward extraction against TRL's internal DPO reward computation on a batch of trained-on pairs, on five points: (i) our tokenization matches TRL's token-for-token; (ii) our per-pair values agree with TRL's logged ones up to a small residual from batching and floating point (TRL's padded batched forward passes versus our unpadded per-example ones); (iii) the reward is exactly linear in β, ruling out a scaling bug; (iv) an unmasked-prompt control shifts it far more than that residual, ruling out a masking bug; (v) π_{ref} is bit-identical to TRL's adapter-disabled reference. The residual's source is not characterized, but it cannot affect the reported metrics: pairwise accuracy depends only on the *sign* of each scorer's own difference.

Bringing up the training pipeline surfaced four correctness bugs, none of which crashed: all would have produced plausible, wrong results (Table 4).

Table 4: Correctness bugs found and fixed before trusting any result. The first two fixes are why the SFT adapter is merged into standalone weights instead of being left live beneath DPO/reward-model training.

Bug	Symptom if unfixed	Fix	Verified by
Wrong π_{ref}	DPO trains against the raw base model, not the SFT policy	Merge the SFT adapter into the base weights before DPO/RM	DPO loss is $\ln 2$ at step 0[1]
Frozen RM head	Scalar head stays at random init; training does nothing to it	Merge SFT first, then attach a fresh sequence-classification head	`trainable == True` (measured)
Train-pair mismatch	RewardTrainer drops long pairs, DPOTrainer truncates them: 7,520 vs. 8,000 pairs at the same nominal budget	Draw both trainers from one common length-eligible pool	Pool hash-identical between runs
Multi-GPU batch inflation	HF `Trainer` auto-wraps in `DataParallel`; effective batch silently $4\times$ larger	Pin every run to a single GPU	n/a

3.3 H1: in-distribution comparison

In distribution the explicit reward model wins (Table 5) in every individual seed, and all five seed-level McNemar tests are significant ($p < 0.0001$ to 0.002). Those tests run on the full test set (§2.4), so they corroborate the ranking without testing the length-controlled column itself. Here the two agree, the explicit model leading on both (0.691 vs. 0.632 raw). **H1 is supported.**

[1] A zero-init LoRA gives $\pi_\theta = \pi_{\text{ref}}$, hence zero margin and $-\log\sigma(0) = \ln 2$. The first *logged* value is 0.679, TRL's mean over the first 20 steps.

Table 5: Length-controlled pairwise accuracy (mean ± 95% CI, 5 seeds). "Diff." is explicit minus implicit, paired within each seed.[2] "95% CI excludes/includes 0" is defined in §2.4; a negative, CI-excludes-0 difference means the implicit reward is ahead.

Test set	DPO implicit	Explicit RM	SFT baseline	Diff. (E−I)	95% CI
UltraFeedback (ID)	0.600 ± 0.026	0.663 ± 0.009	0.529	$+0.062 \pm 0.030$	excl. 0
RewardBench: Chat	0.800 ± 0.044	0.747 ± 0.080	0.500	-0.053 ± 0.058	incl. 0
RewardBench: Chat-Hard	0.535 ± 0.052	0.615 ± 0.061	0.661	$+0.079 \pm 0.109$	incl. 0
RewardBench: Safety	0.492 ± 0.036	0.549 ± 0.059	0.530	$+0.058 \pm 0.074$	incl. 0
RewardBench: Reasoning	0.850 ± 0.003	0.663 ± 0.019	0.906	-0.187 ± 0.020	excl. 0
HH-RLHF (harmless)	0.472 ± 0.016	0.429 ± 0.036	0.435	-0.043 ± 0.030	excl. 0

3.4 H2: does the gap widen out of distribution?

Pairing the explicit-minus-implicit difference *within* each seed, so run-to-run variance cancels, three of five OOD sets show no resolvable difference (Chat, Chat-Hard, Safety). On the two that do, Reasoning and HH-RLHF, the difference runs opposite to H2, favoring the implicit reward (Figure 3, Table 5). **H2 is not supported.**

Chat is the one set where the two statistics disagree. Its McNemar tests are all significant, since on the full 358 pairs the explicit model leads 0.904 to 0.746. But length control keeps only 32 pairs, and there the ranking reverses to 0.747 against 0.800 with an interval spanning zero. Each reading is correct about its own metric, and which to believe turns on how much of that lead is the length cue pick-longer exploits to 0.805 on this very subset (Table 3), a question §3.6 returns to.

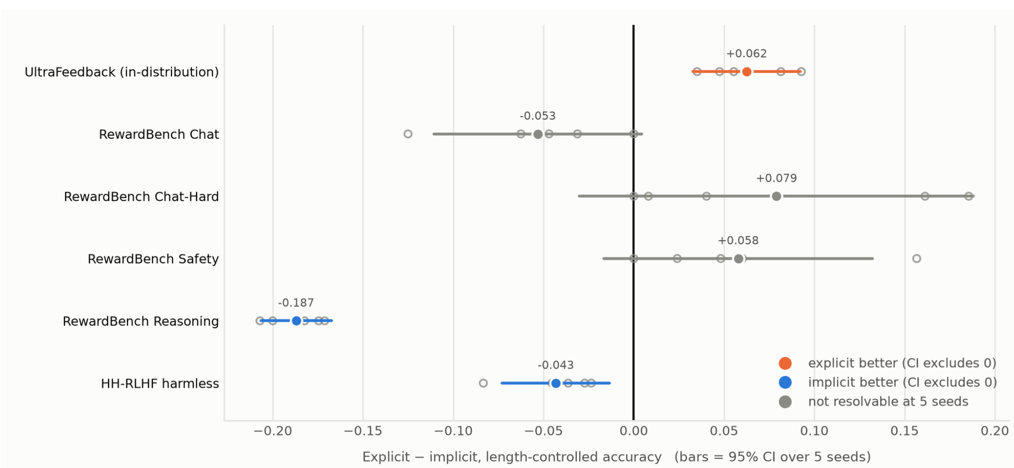

Figure 3: Explicit minus implicit length-controlled accuracy, paired within each seed; bars crossing zero (grey) are not resolvable.

3.5 An unplanned finding: reward training can underperform the SFT baseline

Scored against the SFT log-probability baseline instead of chance, the baseline beats *both* trained scorers on RewardBench-Reasoning, and beats the implicit reward resolvably on Chat-Hard and Safety, where the explicit model's gap is not resolvable (Figure 4). On three of the five shifted sets,

[2] Computed from unrounded per-seed values, so it can differ by up to 0.001 from the rounded means shown.

then, reward-model training left at least one scorer worse at ranking than the SFT policy's own untuned log-probability, and not on the sets we would have guessed. Reasoning, the only subset where the baseline beats both, is RewardBench's code-and-mathematics split, not its adversarial one; on Chat-Hard, built specifically to invert the easy cues, the explicit model holds its ground.

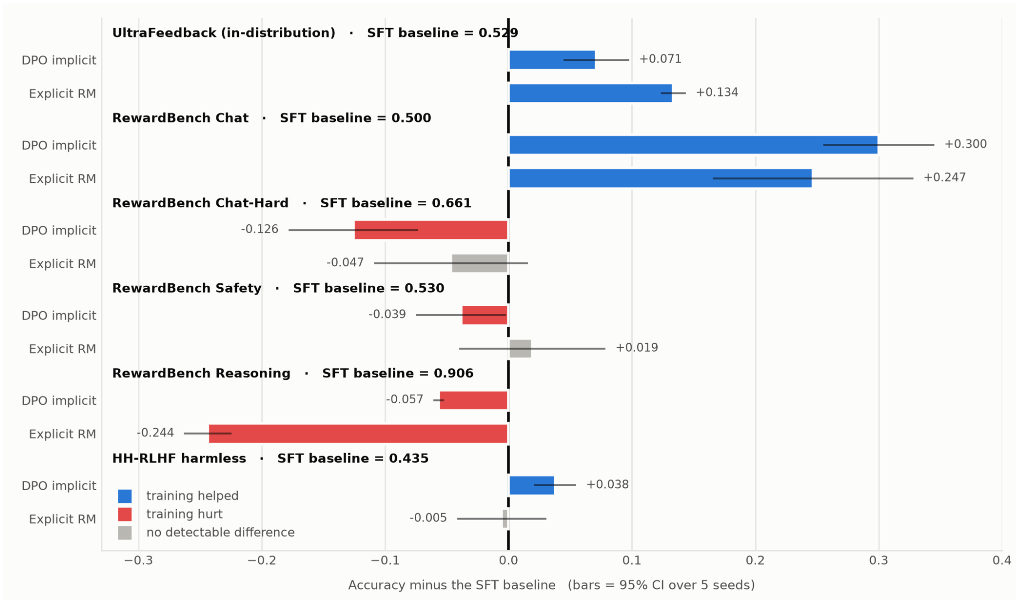

Figure 4: Accuracy minus the SFT log-probability baseline; bars left of zero mean reward training left the scorer *worse* at ranking.

3.6 Length confound and surface-feature associations

Length control matters: the explicit reward model scores below chance on Chat-Hard before it (0.414) and recovers once length-matched (Table 5), a shift large enough to flip its ranking relative to chance. More generally, the explicit model and the SFT baseline both pick longer, more formatted, more numeric responses more often than the dataset labels do, by 3 to 16 points depending on the feature and by 12 and 16 points on length itself. The implicit reward deviates less on average and in the opposite direction, picking those same features *less* often than the labels (Figure 5). It is not uniformly closer, though: its largest deviation anywhere in the figure is on hedging and disclaimers (-12.6 points), where the explicit model is much closer. Response length is therefore one likely contributor to H1's in-distribution gap and to the explicit model's raw-accuracy collapse on length-inverted subsets like Chat-Hard.

Table 5 also holds one result length alone does not explain. On HH-RLHF every scorer, trained or not, ranks below chance on the length-matched pairs (0.472, 0.429, 0.435). Length is part of it: pick-longer scores 0.445 on the full set (Table 3) and the explicit model and SFT baseline score 0.446 and 0.451, tracking that floor almost exactly, so the shorter response is usually the labelled one and any scorer preferring length inherits a below-chance ranking. But length matching removes that cue and all three stay below chance. HH-RLHF is also the only set with human labels, and its "harmless" preferences reward refusal and de-escalation, which nothing in UltraFeedback training rewards. We flag this as unexplained.

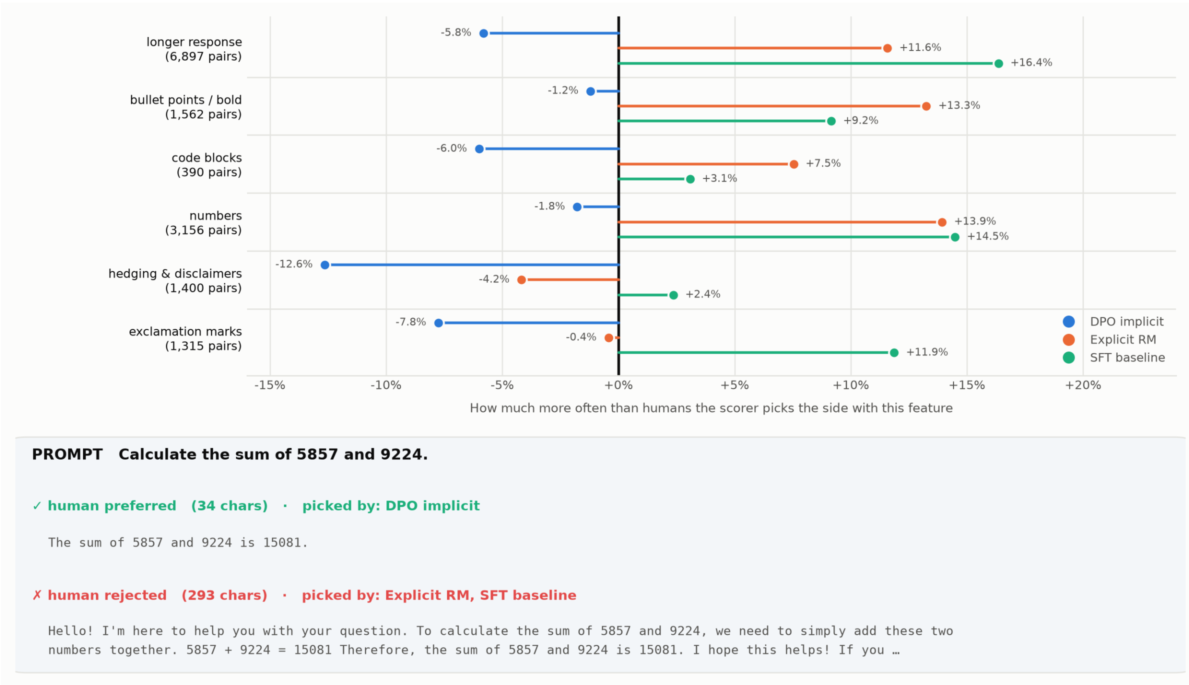

Figure 5: Deviation from each dataset's own label rate, per surface feature (top), with one worked example (bottom). The figure reads "human", but UltraFeedback's labels are an AI judge's (§2.2).

3.7 Ablations: extended training and length-bias drift

Re-running both conditions for four epochs (two seeds), with held-out pairwise accuracy measured every 100 steps on a shared slice of 400 UltraFeedback pairs, tests whether the one-epoch ranking is an artifact of stopping two still-improving models (Figure 6). Both trainers get the same slice, though the reward-model trainer drops a few over-length pairs, so its own evaluation set is 394–398 of the 400. The explicit reward model climbs to a peak of 0.731 near epoch 2 and then overfits; the implicit reward is flat from the start, peaking at 0.676 near epoch 1. The explicit model is ahead at every epoch past $\sim$0.4, by ≈ 0.048 at the one-epoch mark (0.724 vs. 0.676) and 0.055 at each method's own best checkpoint: the gap widens slightly over the run, it does not close.[3] **H1 remains supported through the full four-epoch run; the ranking does not invert.** We stop short of calling this "convergence": the reward model is still visibly overfitting by epoch 4, and training loss keeps falling long after held-out loss turns (Figure 7), so loss is not a usable stopping signal here. Its best checkpoint is near epoch 2, which makes four epochs worse than one for it specifically.

[3] Different metric from Table 5's +0.062, and measured on a 400-pair subset of that test set, so the two are not directly comparable.

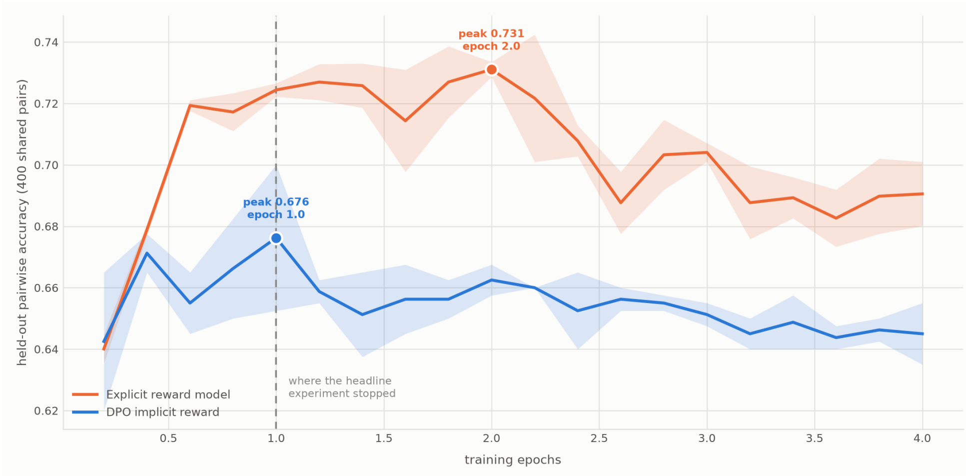

Figure 6: Held-out pairwise accuracy across four training epochs; the band spans two seeds.

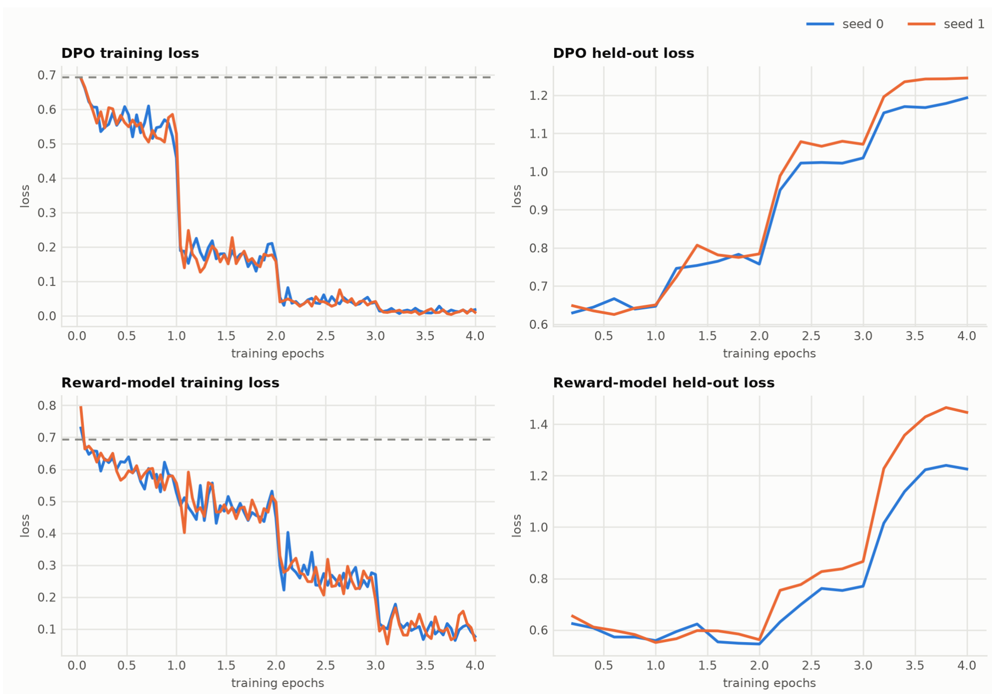

Figure 7: Training loss (left) against held-out loss (right) across the four-epoch run.

Scoring all eight saved checkpoints of each condition on a separate 499-pair draw (492 with a length difference to measure) tests whether either scorer's length bias grows with training (Figure 8). The hypothesis is **wrong for the explicit model**: its bias peaks early (+0.069 against the label rate at epoch 1.5) and fades to +0.002 by epoch 4, on roughly the schedule of its accuracy peaking and falling (0.719 at epoch 1.5, 0.685 at epoch 4 on this draw). The implicit reward drifts the other way, from −0.134 to −0.232, ending up picking the *longer* response only 32.5% of the time where the labels pick it 55.7%, with accuracy falling from 0.649 at epoch 1 to 0.573 by epoch 4. This is consistent with an anti-length preference induced by DPO training that contributes to why the implicit reward saturates early and then degrades. Two caveats: both drifts are single-seed, and this draw is not the 400-pair slice behind Figure 6, so accuracies here sit a point or two off that curve.

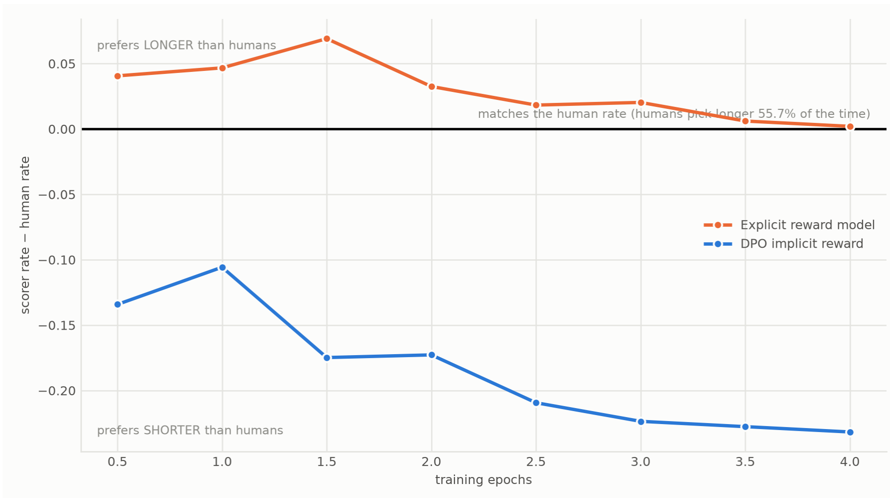

Figure 8: Length preference relative to the label rate, across eight checkpoints per model. Measured on UltraFeedback `test_prefs`: as in Figure 5, the axis reads "human" but the labels are an AI judge's.

3.8 Limitations

Several limitations bound how far these results generalize.

Sample size. The three sets marked † in Table 1 fall below 100 length-matched pairs, the filter having removed roughly 90% of each. Their length-controlled numbers, given to three decimals only for consistency with the rest of the table, indicate direction and not precise estimates; a length-adjusted regression would recover the power the filter costs.

What is not matched. Wall-clock time and FLOPs are not, since DPO performs an extra frozen-reference forward pass every step and took about 1.7× the reward model's wall-clock time for the same optimizer steps. This is a matched *training budget* (Figure 1), not matched compute in the FLOP sense.

π_{ref} **was trained once.** Seeds vary only the second-stage (DPO/reward-model) training, so the intervals capture variance from that stage given one fixed reference, not of the method as a whole. It is also why the SFT baseline's interval in Figure 4 has zero width.

Associations, not mechanisms. The surface-feature effects in Figure 5 are correlational and overlapping, since longer responses also carry more numbers and bullets, so isolating length would need a joint model. Likewise the readings in §3.5 and §3.7 of *why* training degrades performance are consistent with the data, not established causes.

Calibration is poor across the board (ECE 0.22–0.45), which matters if the practical pitch is reusing either scorer as a probability estimate and not only as a ranker.

Scope. Every result is at one model scale (0.5B), one β (0.1), and one training budget (8,000 pairs); the $\beta \in \{0.05, 0.1, 0.3, 0.5\}$ and budget $\in \{2\text{k}, 8\text{k}, 32\text{k}\}$ ablations planned in the proposal remain future work.

4 Conclusion

Budget-matched from one shared starting checkpoint (Figure 1), an explicitly trained Bradley–Terry reward model beats DPO's implicit reward on held-out, in-distribution preference pairs, and the ranking holds through an extended four-epoch run without inverting (**H1 supported**). The predicted widening of that gap under distribution shift does not occur. Where a difference is resolvable, it favors the implicit reward instead (**H2 not supported**). Unplanned, an SFT log-probability baseline with no reward-specific training beats both trained scorers on RewardBench-Reasoning, and beats the implicit reward on Chat-Hard and Safety as well. DPO's "secretly a reward model" claim therefore holds only in a narrow, in-distribution sense at this scale: an explicit reward model is measurably better in distribution, but under shift neither trained scorer consistently beats the untrained baseline.

Three limitations bound it (§3.8 has the full list): every number is at one model scale, one β, and one training budget, so we cannot claim the ranking is stable along any of those axes; three of the five shifted sets keep under 100 pairs after length matching, which is why so many OOD differences are unresolvable; and π_{ref} was trained once, so the intervals describe variance in the second training stage, not in the method. Future work should run the planned β and budget sweeps, re-seed the reference policy, and replace length filtering with a length-adjusted regression.

References

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